

CASE STUDY 3: AI-Based Evaluation System for High-Stakes Examinations

Designing a Controlled, Template-Guided AI Evaluation Framework for Accuracy, Fairness, and Accountability

1. Problem Statement

High-stakes examinations require accurate, fair, and consistent evaluation. Manual grading is slow and inconsistent at scale, while fully automated AI grading introduces unacceptable risks of bias, hallucination, and lack of transparency. A controlled AI-assisted approach is required to improve efficiency without compromising trust.

2. Users & Stakeholders

Primary Users

- Examination authorities and evaluation committees
- Subject matter experts

Stakeholders

- Exam candidates
- Regulatory and oversight bodies

3. Solution Options Considered

Option 1: Fully Manual Evaluation — Rejected

Accurate but slow, expensive, and difficult to scale.

Option 2: Fully Automated AI Evaluation — Rejected

High risk of bias and lack of explainability for high-stakes decisions.

Option 3: Template-Guided AI-Assisted Evaluation — Selected

AI assists evaluators by mapping responses to predefined templates, while humans retain final scoring authority.

4. High-Level Architecture & Workflow

1. Digitization and preprocessing of responses
2. Mapping responses to structured evaluation templates
3. AI suggests provisional scores and highlights evidence
4. Human evaluators review and finalize scores
5. Audit logs maintained for transparency

5. Responsible AI Risks & Mitigation

- Bias controlled through standardized templates
- Hallucination prevented via template-only AI assistance
- Transparency ensured through scoring rubrics and logs
- Accountability retained by human evaluators

6. Expected Impact & Success Metrics

- Reduced evaluation time
- Improved consistency across evaluators

- Faster result declaration
- Increased trust among stakeholders

7. Conclusion

This case study highlights that AI must be used cautiously in high-stakes contexts. A template-guided, human-in-the-loop approach balances efficiency with fairness and accountability.